

Question

HA is a monoprotic weak acid. The solubility s (mol/L) of the sparingly soluble salt MA in buffer solutions of different acidity is experimentally studied to determine the solubility product constant of MA and the ionization constant of HA. The experimental data are shown in the table below:

pH	$s(10^{-4} \text{ mol/L})$
4.00	2.7
3.70	3.2
3.52	3.5
3.40	3.9
3.22	4.5

Please find the variables with relevant relationships through the equilibrium and conservation relationships in the solution, and obtain the equation by fitting the experimental data, and use the obtained results to find K_{sp} and K_{a} .

A. pH has a linear relationship with s

B. $[\text{H}^+]$ and s have a linear relationship.

C. $K_{\text{sp}} = 4.9 \times 10^{-8}$

D. $K_{\text{a}} = 1.9 \times 10^{-4}$

E. $K_{\text{sp}} = 4.8 \times 10^{-6}$

F. $K_a = 2. \times 10^{-4}$

G. All other options are incorrect.

H. Apart from G and H, there are more than one correct option.

Answer

Correct Answer: **D**

Detailed Explanation

According to the definition of K_{sp} , the following derivation can be made:

$$K_{sp} = [M^+][A^-] = [M^+]c(HA)\delta(A^-) = \frac{[M^+]^2 K_a}{[H^+] + K_a} = \frac{s^2 K_a}{[H^+] + K_a}$$

CHECKPOINT

1 PTS

Derivation yields $K_{sp} = \frac{s^2 K_a}{[H^+] + K_a}$ or equivalent form

After rearrangement, the linear equation is obtained

$$s^2 = \frac{K_{sp}}{K_a} [H^+] + K_{sp}$$

CHECKPOINT

1 PTS

The linear equation is $s^2 = \frac{K_{sp}}{K_a} [H^+] + K_{sp}$

Therefore, experimental data can be substituted for linear regression, yielding $s^2 = 2.6 \times 10^{-4} [H^+] + 4.8 \times 10^{-8}$

CHECKPOINT

1 PTS

The slope of the linear equation is 2.6×10^{-4} , and the intercept is 4.8×10^{-8}

That is, $\frac{K_{sp}}{K_a} = 2.6 \times 10^{-4}$, $K_{sp} = 4.8 \times 10^{-8}$, therefore $K_a = 1.9 \times 10^{-4}$

CHECKPOINT

1 PTS

$$K_{sp} = 4.8 \times 10^{-8}, K_a = 1.9 \times 10^{-4}$$